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Tutorial Sheet
Hashing

Q.1) Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function x mod 10. Which of the following statements are true?

1. 9679, 1989, 4199 hash to the same value
2. 1471, 6171 hash to the same value
3. All elements hash to the same value
4. Each element hashes to a different value

a) 1 only
b) 2 only
c) 1 and 2 only
d) 3 and 4 only

Q.2) Consider the hash table of size seven, with starting index zero, and a hash function (3x+4) mod 7. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that - denotes an empty location in the table

a) 8, - , - , - , - , - , 10
b) 1, 8 , 10 , - , - , - , 3
c) 1, - , - , - , - , - , 3
d) 1, 10 , 8 , - , - , - , 3

Q.3) Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?

a) $(97 * 97 * 97) / 100^3$
b) $(99 * 98 * 97) / 100^3$
c) $(97 * 96 * 95) / 100^3$
d) $(97 * 96 * 95) /(3! * 100^3)$

Q.4) Consider a hash table P[0, 1 ,...,10] that is initially empty. The hash maintained using open addressing with linear probing. The hash function used is h(x) = (x + 7) mod 11. Consider the following sequence of insertions performed on P: 1, 13, 22, 15, 11, 24
Which of the following positions in the hash table is/are empty after these insertions are performed?

a) 0
b) 10
c) 2
d) 1

Q.5) In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the ith probe in the open address table is $[h_1(k) + i*h_2(k)] \bmod m$. The following keys are inserted in the given order. 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____(Answer in integer).

Q.6) The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function h(k) = k mod 9. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____(Answer in integer)

Q.7) The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function h(k) = k mod 10 and linear probing. What is the resultant hash table?

Q.8) Given a hash table T with 25 slots that stores 2000 elements, the load factor for T is ____?

Q.9) A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

Q.10) Consider a hash table with 9 slots. The hash function is h(k)=k mod 9. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are?

Answer key

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
c	b	a	c	10	3	-	80	6	-